Viridis: A Low-Power, Linux-Powered Super Server!

Viridis is a 5W server! It is interesting to note that Linux was the first choice of operating system for this unique high-performance computing system, and proved to be the best.

One of the biggest problems faced by data-centre operators is the amount of power consumed and dissipated by the servers. As a result, there is a lot of focus on building low-power servers these days. ARM processors are known for their low-power capabilities, but have traditionally been used more in mobile devices and laptops than in supercomputers. However, in the last couple of years, there has been a lot of interest in building ARM-based servers, and such products are being launched by companies like Boston, Marvell, Tilera and Calxeda. One that seems very promising is Boston's Viridis, an energy-efficient data-centre solution based on Calxeda's low-power, ARM-based System-on-Chip (SoC).

ARM—a logical choice for data centres

"In data centres, the server is the revenue-generating asset of the business, and there is a strong business motivation to look at new technology. Companies look to drive down the acquisition cost of the hardware, and the running cost of the platforms. You can think of these data centres as energy-constrained systems. A data centre gets allocated a certain amount of power, and it must generate as much revenue as it can, and minimise running costs. As the servers are specialist machines, there is an opportunity for silicon and systems companies to optimise platforms for a specific application. ARM has proved in the mobile phone space that the way ahead is development of highly integrated SoC devices," says Guru Ganesan, managing director, ARM, India Operations.

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